

Taha Gamal Alieldin

Mechatronics Engineering Graduate

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Summary

Mechatronics Engineering graduate specializing in automation, robotics, and machine learning. Led an 8-member team building an autonomous surveying swarm. Gained industrial experience at ENPPI and Norpetco, with strong communication and problem-solving skills from customer-facing roles.

Professional Experience

ENPPI, Instrumentation & Control Intern Studied instrumentation and process control (sensors, transducers, P&ID) with exposure to SCADA, PLC, and DCS systems.	07/2025
Norpetco (North Petroleum Company), Production & Field Operations Trainee Completed office and on-site field training, observing power generation, artificial lift equipment (Lufkin beam pumps, ESP), and firefighting systems.	07/2023 – 08/2023 Cairo (office) + Fields
Luban Workshop, Ain Shams University, Embedded Robotics Trainee Built and programmed a differential-drive MicroMouse robot.	08/2025
Sales & Receptionist, Fitness Time Gym #1 Sales Representative for three consecutive months, generating ~300K EGP in monthly revenue.	2022 – 2025 Mokattam, Cairo

Certifications & Courses

Advanced Automation, HA Consulting Group Expected Sep 2026.	06/2026 – Present
Basic Automation, HA Consulting Group A+ in all three modules: Classic Control, PLC Basic Programming (ladder logic, Siemens TIA Portal), and Electric Motor & Drive Programming (VFDs).	07/2024 – 09/2024
OSHA Construction & Industry Safety Course, Orascom Construction	02/2025

Education

Ain Shams University, BSc. Mechatronics Engineering GPA: 2.75 (B-) Graduation project grade: A+	09/2021 – 07/2026
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Selected Projects

AI-Enhanced Autonomous Surveying Swarm, Graduation Project Led an 8-member team to build an autonomous surveying swarm on ROS 2 (Humble). Built Voronoi coverage algorithms and Gazebo simulations; pitched the prototype to win a 100K EGP grant from ASRT.	08/2025 – 06/2026
Real-Time Human Behavior Recognition System, Hybrid CNN-LSTM Pipeline for Action Classification Directed a 4-member team building a real-time CNN-LSTM computer-vision system for action recognition, reaching the cohort's highest accuracy.	10/2025 – 01/2026
Furuta Pendulum Control System, LQR & Hardware-in-the-Loop Implementation via MATLAB/Simulink Designed an LQR controller to balance a rotary pendulum against disturbances, modeled in SimMechanics and validated via HIL, with a state-flow controller.	10/2025 – 01/2026

Technical Skills

- Control & Automation:** PLC (ladder logic), classic control, LQR, state machines.
- Programming & Tools:** MATLAB/Simulink, Python, C++, ROS 2, Gazebo, OpenCV, Automation Studio.
- ML/Vision:** CNN, LSTM, deep learning.
- Mechanical/Design:** Inventor, Ansys.